

Tharindu Lakshan

COMPUTER ENGINEERING UNDERGRADUATE

📍 No:23/92, Western Park, Kalagedihena, Gampaha District, Sri Lanka

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About Me | Portfolio - tharindun.online

Enthusiastic Computer Engineer with a strong foundation in AI, full-stack web development, and cloud-native systems. Experienced in real-world projects and passionate about building scalable, user-focused applications, with a security- and reliability-minded approach and the ability to adapt quickly. Eager to grow by contributing to impactful development work, integrating AI capabilities where they add value.

Education

University of Peradeniya

2022 - 2026

BSc.Eng(Hons.) Specializing in Computer Engineering

GPA : 3.84/4.00 - First Class

Royal College, Colombo 07

2012 - 2020

G.C.E. Advanced Level Examination

3As

Experience

Software Engineer Intern in AI Integration team – WSO2 (Pvt) Ltd.

Apr. 2026 – Present

1. Streaming Support for WSO2 Integrator AI Libraries.

- Designed and tested first-class token-by-token AI streaming across the WSO2 Integrator ecosystem.
- Introduced a **unified streaming abstraction** with vendor-specific implementations for **7 LLM providers**, keeping full backward compatibility with existing non-streaming APIs.

2. Google Gemini Model & Embedding Provider for WSO2 Integrator AI ecosystem 🌐 Provider | 🌐 Tests

- Built a Gemini chat and embedding provider for the framework's model-provider interface.

3. Built AI data loaders for the WSO2 Integrator AI ecosystem. 🌐 Blob | 🌐 File | 🌐 Drive

- Enabled loading of documents from Azure Blob Storage, Azure File Storage, and Google Drive into AI pipelines for RAG and downstream processing.

4. Added parallel tool-calling support across model providers.

- Implemented concurrent tool execution for OpenAI, Azure OpenAI, and Anthropic providers, cutting latency for multi-tool agent workflows.

5. Expanded AI Support. Added Claude Opus 4.7/4.8 support and raised output-token limits across 7+ providers.

- Technologies: Java, LLM, Azure, Embeddings, SSE, RAG, MCP, Git, Apache Tika, Apache POI

Undergraduate Teaching Assistant – Dept. of Computer Engineering, University of Peradeniya 2023 – 2025

- Served as a Casual Instructor for **CO1010** – Programming for Engineers I, **GP106** – Computing, **CO225** – Software Construction, and **CO322** – Data Structures & Algorithms.

Research

1. Security Testing and Threat prediction for Wi-Fi 7 MLO - Group | 🌐 Repo | 🌐 Web Page

- Analyzing the Wi-Fi 7 MAC **backoff mechanism** across different counter states to uncover protocol-level vulnerabilities.
- Identifying potential attack surfaces in **Multi-Link Operation (MLO)** and designing mitigation strategies to strengthen overall Wi-Fi 7 security.
- Contribution: Research gap analysis, GNN-based threat prediction implementation, and dashboard design.
- Technologies: ns-3, C++, Python, Kafka, TimescaleDB, Grafana, Docker, Containerlab

Projects

1. LIGHTIFY – Smart Home Lighting System - Group | 🌐 Repo | 🌐 Web Page

- Intelligent home lighting system** that detects room occupancy via sensors and dynamically adjusts lighting, with a **Spring Boot** backend and **MongoDB** deployed on **AWS** for real-time synchronization.
- Used **ESP32** embedded systems with **MQTT** over **AWS IoT Core** for reliable, real-time device control between the lighting circuits and mobile app.

- Contribution: Built the React Native mobile app and selected Spring Boot backend features, and designed and assembled the hardware.
 - Technologies: React Native, Spring Boot, AWS IoT Core, MongoDB, ESP32, MQTT
2. **Smart Parking Management System – Cloud-Native Microservices Platform - Individual** |  Repo
- A **cloud-native platform** for drivers to find, reserve, and pay for parking on a live map, with real-time availability, overstay alerts, and an operator analytics dashboard for occupancy and revenue.
 - Engineered as an **event-driven system** of 10 Spring Boot microservices over Kafka, with a choreographed reservation→payment Saga, geospatial search, WebSocket updates, and CI/CD to Oracle Cloud.
 - Technologies: Java 21, Spring Boot 3.3, Spring Cloud Gateway, Apache Kafka, PostgreSQL/PostGIS, Redis, React 18, Flutter, Docker, GitHub Actions, Terraform, Prometheus/Grafana, JWT, Oracle cloud
3. **Lakshan Home Collection – E-commerce website - Individual** |  Repo
- **Responsive full-stack e-commerce site** with product search and filtering, built with React 18, Redux Toolkit, and Tailwind CSS
 - Spring Boot backend with MongoDB Atlas, JWT authentication, and role-based access control, supporting product management and product uploads; deployed to AWS EC2 via a GitHub Actions CI/CD pipeline.
 - Technologies: React 18+, Redux Toolkit, Spring Boot 3.x, MongoDB Atlas, JWT, Tailwind CSS, shadcn/ui, GitHub Actions, AWS EC2
4. **CardioGuard – Cardiac Patient Monitoring System - Group** |  Repo |  Web Page
- **Cardiac patient database and health monitoring system** powered by **machine learning** to assess heart-failure probability and overall cardiac risk, built with **Spring Boot** and **React**.
 - Secure **JWT-based**, role-specific dashboards for doctors, patients, and admins.
 - Contribution: Developed features across both the **React** frontend and **Spring Boot** backend.
 - Technologies: Java, Spring Boot, React, Python, Tailwind CSS, Machine Learning

Technical Skills

- **AI & LLM:** LLM APIs (OpenAI, Anthropic, Gemini, Azure), RAG, MCP, SSE Streaming, Prompt Engineering
- **Cloud & DevOps:** AWS (S3, EC2, IoT Core), Oracle Cloud, Vercel, Docker, GitHub Actions, Terraform, MQTT
- **Languages:** Java, Python, JavaScript/TypeScript, C
- **Web & Backend:** React, React Native, HTML, CSS, Tailwind CSS, Spring Boot
- **Databases:** MySQL, PostgreSQL, MongoDB
- **ML & Data:** TensorFlow, NumPy, OpenCV, Matplotlib, Logistic Regression, Neural Networks
- **Concepts:** Object-Oriented Programming, Data Structures & Algorithms, Network Security, Embedded Systems
- **Tools & Version Control:** Git, GitHub, VS Code, IntelliJ, Postman, Android Studio, Cisco Packet Tracer, Wireshark

Extracurricular Activities

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| 1. Assistant Secretary – Hackers' Club, University of Peradeniya | 2023–2024 |
| Organized the PreCoders 2024 6-hour coding competition and competed in the event. | |
| 2. Member – Rotaract Club, University of Peradeniya | 2023–Present |
| 3. Member – AeroGenz Club, University of Peradeniya | 2024–2025 |

Certificates

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| ▪ AWS Cloud Practitioner Essentials - (AWS) | ▪ JavaScript Basics - Coursera |
| ▪ Fundamentals of Java Programming - Coursera | ▪ Programming for Everybody (Python) - Coursera |
| ▪ Python Data Structures - Coursera | ▪ AI/ML Engineer – Stage 1 and 2 - SLIIT UNI |
| ▪ C for Everyone: Programming Fundamentals - Coursera | ▪ Introduction to Cybersecurity - CISCO |

References

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| ▪ Sasindu Alahakoon – Associate Technical Lead,
WSO2 Pvt Ltd
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sasindu@wso2.com +94 76 590 6142 | ▪ Dr. Isuru Nawinne – Senior Lecturer
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